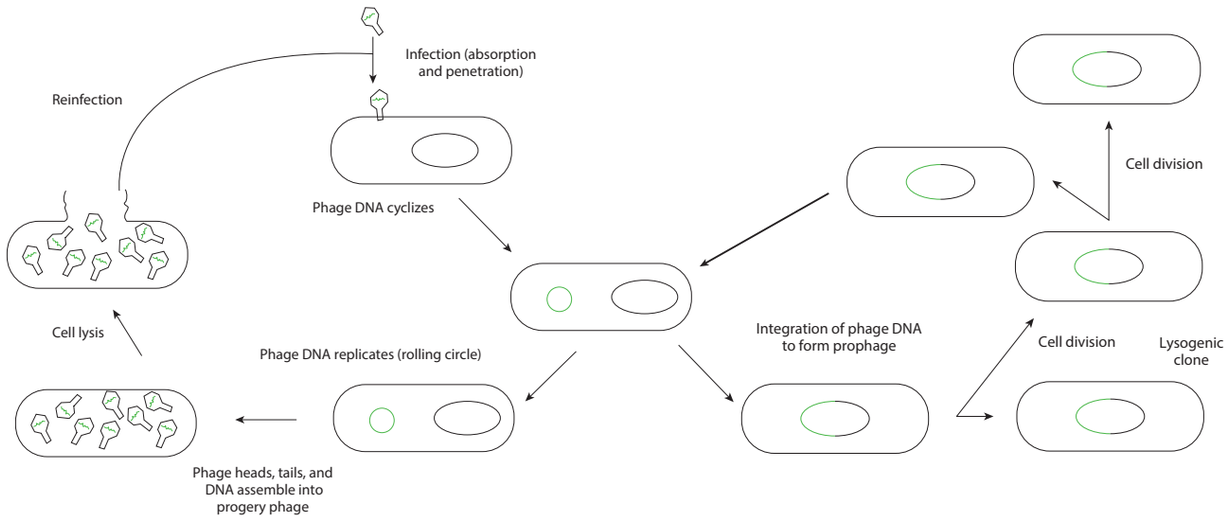


Figure 12.20: Induction of lysogen

Sometimes, bacterial genes are incorporated into a phage capsid because of errors made during the virus life cycle. The virus containing these genes then infects them into another bacterium, resulting in the transfer of genes from one bacterium to the other. Transduction may be the most common mechanism for gene exchange and recombination in bacteria.

There are two very different kinds of transduction.

1. Generalized transduction
2. Specialized transduction

Generalized transduction (Figure 12.21a) occurs during the lytic cycle of virulent and temperate phages. During the assembly stage, when the viral chromosomes are packaged into protein capsids, random fragments of the partially degraded bacterial chromosome also may be packaged by mistake. The resulting virus particles often injects the DNA into another bacterial cell but does not initiate a lytic cycle. Thus in generalized transduction any part of the bacterial chromosome can be transferred. Once the DNA has been injected it may integrate into the recipient cell's chromosome to preserve the transferred genes. About 70 to 90% of the transferred DNA is not integrated but is often able to survive and express itself. However, if the transferred DNA is degraded gene transfer is unsuccessful.

Specialized Transduction (Figure 12.21b) is also called restricted transduction in which only specific portions of the bacterial genome is carried by the phage. When a prophage is induced to leave the host chromosome, exicision is sometimes carried out improperly. The resulting phage genome contains portions of the bacterial chromosome next to the integration site. When this phage infects another bacterium, it transfers the bacterial genes from the donor bacterium along with phage DNA. Here only the bacterial genes that are close to the site of prophage are transferred. So, this transduction is called specialized.

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